

# Supplementary Information

## Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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This document provides the reproducibility, parameter-grid, metric-formula, and complete eight-design comparison material referred to in the main text as “online supplementary material.” All numeric values below were verified directly against the simulation output files `results/six_design_manuscript/six_design_summary.txt`, `results/six_design_manuscript/dissertation_results_extract.txt`, and `results/eight_design_supplementary/eight_design_summary.txt` (the last generated fresh for this document by `code/14_manuscript_supplement_figures.R`). No number below is asserted without a corresponding entry in one of these files.

Unless otherwise noted, all sections other than Section [S6](#) present the **six treatment assignment designs retained in the main text** (Checkerboard, High Incidence Focus, Isolation Buffer, 2×2 Blocking, Balanced Quartiles, and Incidence-Guided Saturation Quadrants), ordered from best (lowest mean squared error, MSE) to worst wherever a ranking is shown. Section [S6](#) is the sole section presenting all eight designs evaluated in the underlying simulation study, including the two variants (Saturation Quadrants, Balanced Halves) dropped from the main comparison as statistically redundant with a retained design.

## S1 Reproducibility and Code Availability

All simulation, estimation, and figure-generation code is organized as a numbered pipeline in `code/`, sourced in dependency order:

| Script                                  | Role                                                                            |
|-----------------------------------------|---------------------------------------------------------------------------------|
| 01_spatial_setup.R                      | 10×10 grid construction; rook/queen spatial weight matrices                     |
| 02_incidence_generation.R               | Baseline incidence generation (iid, spatial, Poisson modes)                     |
| 03_designs.R                            | The eight treatment assignment designs (Section <a href="#">S5</a> )            |
| 04_estimation.R                         | DIM and MLE ( <code>lagsarlm</code> ) estimation of $\tau$ , with CI extraction |
| 05_run_simulation.R                     | Main orchestrator: parameter grid, seeding, simulation loop                     |
| 06_visualizations.R                     | Plotting and tabulation functions                                               |
| 10_statistical_comparisons.R            | Friedman/Nemenyi/Wilcoxon hypothesis testing                                    |
| 12_six_design_statistical_comparisons.R | Six-design re-analysis (statistical redundancy check)                           |
| 14_manuscript_supplement_figures.R      | Figures and tables for this document                                            |

## Seeding protocol

Every simulated scenario uses a distinct, deterministic seed rather than a single global seed for the whole run, so that any individual scenario can be reproduced in isolation and re-ordering scenarios does not change their results. The seed for scenario (`inc_mode`,  $\rho_X$ , `nb_type`,  $\rho$ ,  $\gamma$ , `spill_type`, `design_id`,  $\tau$ ) is

```
scenario_seed <- digest::digest2int(paste(inc_mode, rho_x, nb_type, rho,
    gamma_val, spill_type, d_id, true_tau, sep = "|"))
```

followed by `set.seed(scenario_seed)` immediately before that scenario’s treatment-assignment and outcome draws. Baseline incidence  $X$  is generated *once* per (incidence mode,  $\rho_X$ ) configuration — not per scenario — and held fixed across all downstream (`nb_type`,  $\rho$ ,  $\gamma$ , `spill_type`, `design_id`,  $\tau$ ) combinations built on top of it, matching how a trial team would use a single historical incidence surface rather than redrawing it for every design comparison. Only the first outcome-resample column of that incidence matrix (the “historically observed” incidence) is made available to incidence-aware designs (High Incidence Focus, Balanced Quartiles, Balanced Halves, Saturation Quadrants, Incidence-Guided Saturation Quadrants); the remaining columns are used only to generate outcome noise.

## S2 Full Parameter Grid

Table S1: Full simulation parameter grid. The eight-design comparison (Section S6) uses all 8 design levels; the main-text six-design comparison uses the 6 retained levels.

| Parameter                         | Values                                                                                      | Levels   |
|-----------------------------------|---------------------------------------------------------------------------------------------|----------|
| True treatment effect $\tau$      | 0.8, 1.0, 1.5, 2.0, 3.0                                                                     | 5        |
| Incidence configuration           | iid (1) + spatial ( $\rho_X \in \{0.20, 0.50\}$ ) + Poisson ( $\rho_X \in \{0.20, 0.50\}$ ) | 5        |
| Neighbor definition               | rook, queen                                                                                 | 2        |
| Outcome spatial dependence $\rho$ | 0.00, 0.01, 0.20, 0.50                                                                      | 4        |
| Spillover magnitude $\gamma$      | 0.5, 0.6, 0.7, 0.8                                                                          | 4        |
| Spillover exposure regime         | control_only, both                                                                          | 2        |
| Design                            | 8 designs (Section S5)                                                                      | 8 (or 6) |

Total scenarios:  $5 \times 2 \times 4 \times 4 \times 2 \times 8 \times 5 = \mathbf{12,800}$  for the eight-design comparison, and  $5 \times 2 \times 4 \times 4 \times 2 \times 6 \times 5 = \mathbf{9,600}$  for the six-design comparison used in the main text. At a fixed  $\tau$ , this is 2,560 scenarios (eight designs) or 1,920 scenarios (six designs); within a single incidence configuration at fixed  $\tau$ , there are 320 “blocks” (unique combinations of neighbor type,  $\rho$ ,  $\gamma$ , and spillover regime), which is the blocking structure exploited by the Friedman/Nemenyi tests in Section S6 and throughout the main text.

Checkerboard and High Incidence Focus produce the identical treatment vector  $Z$  for every draw given a fixed incidence surface (`is_design_deterministic()` in `03_designs.R` returns `TRUE` for design IDs 1 and 2). Rather than skipping the redundant draws, `05_run_simulation.R` replicates the single  $Z$  vector across all  $N_{\text{design}} = 25$  design-resample slots and still runs the full outer loop, so the reported  $N_{\text{Valid\_Est}} = 250$  for these two designs as well; because both  $Z$  and the outcome-noise

Table S2: Resample counts per scenario (MLE estimation mode, used throughout this paper).  $N_{\text{design}}$  = number of treatment-assignment draws;  $N_{\text{outcome}}$  = number of outcome-noise draws per assignment draw. Total iterations =  $N_{\text{design}} \times N_{\text{outcome}}$  for both design types; see note below on what the deterministic-design total represents.

| Design type                                        | $N_{\text{design}}$   | $N_{\text{outcome}}$ | Total iterations |
|----------------------------------------------------|-----------------------|----------------------|------------------|
| Stochastic (4 of the 6 retained designs)           | 25                    | 10                   | 250              |
| Deterministic (Checkerboard, High Incidence Focus) | 25 (identical copies) | 10                   | 250              |

draws (`Epsilon.matrix`) are fixed before that loop begins, the 25 design-resample repetitions are exact duplicates of the same 10 outcome realizations rather than 250 independent draws. This does not bias Bias, SD, or MSE (duplicating a sample leaves its mean and variance unchanged) but means the two deterministic designs' effective Monte Carlo sample size is the same 10 independent outcome draws used by every other design, not 250; the Monte Carlo SE formulas in Section S3 therefore mildly understate those two designs' true Monte Carlo uncertainty. This has no bearing on the main text's conclusions, which rest on Checkerboard performing far worse, not on any precision advantage from its larger nominal  $N_{\text{Valid.Est}}$ .

The main text's DIM baseline (used only as an internal validation check during development, never as a scientific comparator in this manuscript) instead used  $N_{\text{design}} = 25$ ,  $N_{\text{outcome}} = 100$  (2,500 total iterations per scenario), reflecting DIM's much lower per-iteration computational cost relative to `lagsarlm`.

### S3 Outcome Metrics

For a scenario with  $N_{\text{valid}}$  non-degenerate estimates  $\{\hat{\tau}_1, \dots, \hat{\tau}_{N_{\text{valid}}}\}$  (iterations excluded only when treatment assignment degenerates to all-treated or all-control, or when `lagsarlm` fails to converge) and 95% confidence intervals  $[\text{CI}_{\ell,i}, \text{CI}_{u,i}]$ :

$$\begin{aligned}
\text{Bias} &= \bar{\hat{\tau}} - \tau, & \bar{\hat{\tau}} &= \frac{1}{N_{\text{valid}}} \sum_i \hat{\tau}_i \\
\text{SD} &= \sqrt{\frac{1}{N_{\text{valid}} - 1} \sum_i (\hat{\tau}_i - \bar{\hat{\tau}})^2} \\
\text{MSE} &= \text{Bias}^2 + \text{SD}^2 \\
\text{Coverage} &= \frac{1}{N_{\text{valid}}} \sum_i \mathbb{I}\{\text{CI}_{\ell,i} \leq \tau \leq \text{CI}_{u,i}\} \\
\text{Power} &= \frac{1}{N_{\text{valid}}} \sum_i \mathbb{I}\{\text{CI}_{\ell,i} > 0\} & (\text{probability of rejecting } H_0 : \tau = 0) \\
\text{Fail\_Rate} &= 1 - \frac{N_{\text{valid}}}{N_{\text{design}} \times N_{\text{outcome}}} \\
N_{\text{Valid.Est}} &= N_{\text{valid}}
\end{aligned}$$

Monte Carlo standard errors (used to gauge simulation precision, not reported as primary

results) are computed under an approximately normal estimator distribution:

$$\begin{aligned} \text{SE}_{\text{Bias}} &= \text{SD} / \sqrt{N_{\text{valid}}} \\ \text{SE}_{\text{Coverage}} &= \sqrt{\text{Coverage}(1 - \text{Coverage}) / N_{\text{valid}}} \\ \text{SE}_{\text{MSE}} &= \sqrt{2 \text{SD}^4 + 4 \text{Bias}^2 \text{SD}^2} / \sqrt{N_{\text{valid}}} \end{aligned}$$

Across all 12,800 eight-design tau-sweep scenarios,  $N_{\text{Valid\_Est}} = 250$  (no degenerate assignments or non-convergent fits) and  $\text{Fail\_Rate} = 0$  throughout, so Monte Carlo SEs are small and uniform across designs and are not a driver of the reported design differences.

## S4 Estimation Model

The direct treatment effect was estimated with an oracle spatial-lag specification fit by maximum likelihood (`spatialreg::lagsarlm`):

$$Y = \rho^{\text{est}} WY + \tau^{\text{est}} Z + \gamma^{\text{est}} \text{Spill} + \beta^{\text{est}} X + \varepsilon,$$

where `Spill` is the true spillover exposure term used in the outcome-generating process ( $WZ$  for the “both” regime,  $WZ \odot (1 - Z)$  for the “control\_only” regime), supplied to the estimator as a known covariate. Because the estimator is given the correct spillover exposure rather than having to infer it, this is a best-case, oracle bound on achievable performance (main text Discussion). The 95% confidence interval for  $\tau^{\text{est}}$  is  $\hat{\tau} \pm 1.96 \times \text{SE}(\hat{\tau})$ , using the model-based standard error from the `lagsarlm` summary object.

### Non-oracle variant (not run for this manuscript)

`estimate_tau()` (`04_estimation.R`) exposes an `include_spill_covariate` toggle; setting it to `FALSE` fits  $Y = \rho^{\text{est}} WY + \tau^{\text{est}} Z + \beta^{\text{est}} X + \varepsilon$  without the spillover covariate, forcing the model to absorb any spillover-induced correlation into  $\rho^{\text{est}}$  or residual error instead of a dedicated term. This non-oracle configuration was not run for this paper; it is listed as priority future work in the main text Discussion, since it represents the estimation scenario a real trial team would actually face (spillover exposure is rarely known with certainty in advance).

## S5 The Eight Treatment Assignment Designs

Checkerboard and High Incidence Focus are deterministic given the incidence surface (the deterministic-design check in `03_designs.R` returns `TRUE` only for design IDs 1 and 2); all other designs are re-randomized independently for each of the  $N_{\text{design}} = 25$  design-resample draws.

## S6 Justification for Design Consolidation: The Complete Eight-Design Comparison

This is the only section in this document (or the main text) presenting all eight designs evaluated in the underlying simulation study. It exists to justify the main text’s decision to drop Saturation Quadrants and Balanced Halves from the six-design comparison, on the grounds that each is statistically indistinguishable from a retained design that implements a closely related mechanism with additional refinement.

Table S3: All eight treatment assignment designs implemented in `03_designs.R`. The two designs marked with  $\dagger$  were evaluated in the underlying simulation study but dropped from the main-text six-design comparison as statistically redundant with a retained design (Section S6).

| Design                                | Implementation                                                                                                                                                                                                                                            | % treated                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Checkerboard                          | Deterministic alternating assignment: cluster $(x, y)$ is treated if $(x + y) \bmod 2 = 0$ , maximizing spatial separation between treated and control clusters.                                                                                          | 50                            |
| High Incidence Focus                  | Deterministic: clusters with observed incidence above the grid median are treated.                                                                                                                                                                        | 50                            |
| Saturation Quadrants $^\dagger$       | The grid is split into four spatial quadrants; each draw randomly permutes the saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ across the four quadrants (independent of incidence), then randomly treats that fraction of clusters within each quadrant.  | $\sim 50$<br>(varies by draw) |
| Isolation Buffer                      | Greedy sequential assignment: clusters are visited in random order; a cluster is treated only if none of its already-assigned neighbors are treated, then it and its neighbors are removed from the pool. Produces sparse, non-adjacent treated clusters. | 20–30                         |
| 2×2 Blocking                          | The grid is partitioned into 2×2 spatial blocks; within each block, exactly 2 of the 4 clusters are randomly treated (1:1 randomization within block).                                                                                                    | 50                            |
| Balanced Quartiles                    | Clusters are stratified into incidence quartiles ( <code>dplyr::ntile</code> ); within each quartile, exactly half are randomly treated.                                                                                                                  | $\sim 50$                     |
| Balanced Halves $^\dagger$            | Clusters are split at the median incidence into two strata; within each stratum, exactly half are randomly treated. A coarser, 2-stratum version of Balanced Quartiles’ 4-stratum logic.                                                                  | $\sim 50$                     |
| Incidence-Guided Saturation Quadrants | As Saturation Quadrants, but the saturation level assigned to each quadrant is rank-ordered by that quadrant’s mean incidence: the highest-incidence quadrant receives 80% treated, the lowest receives 20%.                                              | $\sim 50$                     |

Note that the MSE-based ordering above and the average-Friedman-rank ordering disagree for High Incidence Focus and Isolation Buffer (Table S4): High Incidence Focus has lower mean MSE (0.133 vs. 0.148) but a worse (higher) average rank (5.772 vs. 3.819), because its performance is highly variable across incidence configurations (main text Results) — it wins some scenarios decisively and loses others badly, which pulls its per-scenario rank down even though its pooled mean MSE is competitive.

### Consolidation pair 1: Saturation Quadrants vs. Incidence-Guided Saturation Quadrants

Mean MSE at  $\tau = 1.0$ : Saturation Quadrants = 0.0797, Incidence-Guided Saturation Quadrants = 0.0788 (difference = 0.0008). Nemenyi post-hoc  $p = 1.0000$ ; Wilcoxon signed-rank (Holm-adjusted)  $p = 0.247$ . Neither test detects a significant difference; the two designs are statistically and practically equivalent at this sample size. Incidence-Guided Saturation Quadrants is retained

Table S4: All eight designs ranked by mean MSE at the primary treatment-effect magnitude ( $\tau = 1.0$ ), averaged over the 320 spatial and spillover parameter configurations. Friedman rank-sum test:  $\chi^2(7) = 1091.90$ ,  $p < 2.2 \times 10^{-16}$ . Nemenyi critical difference at  $\alpha = 0.05$ : 0.587; 23 of  $\binom{8}{2} = 28$  pairwise comparisons were significant. Designs marked † were dropped from the main-text six-design comparison.

| Design                                | MSE    | Coverage | Rank (avg. Friedman) |
|---------------------------------------|--------|----------|----------------------|
| Incidence-Guided Saturation Quadrants | 0.0788 | 0.941    | 2.666                |
| Saturation Quadrants†                 | 0.0797 | 0.939    | 2.709                |
| Balanced Quartiles                    | 0.1024 | 0.938    | 3.622                |
| Balanced Halves†                      | 0.1095 | 0.937    | 3.909                |
| High Incidence Focus                  | 0.1330 | 0.966    | 5.772                |
| Isolation Buffer                      | 0.1478 | 0.939    | 3.819                |
| 2×2 Blocking                          | 0.2277 | 0.937    | 6.222                |
| Checkerboard                          | 0.8023 | 0.527    | 7.281                |

because it additionally exploits observed incidence (a strict refinement of the same saturation mechanism), giving it a substantive rationale beyond marginally lower MSE.

### Consolidation pair 2: Balanced Halves vs. Balanced Quartiles

Mean MSE at  $\tau = 1.0$ : Balanced Halves = 0.1095, Balanced Quartiles = 0.1024 (difference = 0.0071). Nemenyi post-hoc  $p = 0.816$  (not significant); Wilcoxon signed-rank (Holm-adjusted)  $p = 1.34 \times 10^{-6}$  (significant). The two tests disagree here: Nemenyi, which compares average ranks across the full 8-design set and is more conservative for closely-clustered designs, does not detect a difference, while the paired Wilcoxon test on the two designs’ raw MSE values directly does. Given the Wilcoxon evidence, the consistently lower MSE, and the finer (4-stratum vs. 2-stratum) incidence stratification, Balanced Quartiles is retained as the representative of this mechanism.

## S7 Design Rankings by Incidence Mechanism

Before conditioning on incidence mechanism, Figures S1 and S2 show the pooled six-design ranking (all incidence configurations combined,  $\tau = 1.0$ ) as critical-difference diagrams (the standard visualization for this kind of repeated-measures comparison; see Demsar 2006, JMLR 7:1–30): designs are positioned by average Friedman rank (Figure S1) or by mean MSE on a log scale (Figure S2), and a bold horizontal bar connects any group of designs that are not significantly different from one another (Nemenyi,  $\alpha = 0.05$ ). Both views show the same two-tier structure among the six retained designs: Incidence-Guided Saturation Quadrants and Balanced Quartiles are statistically indistinguishable from each other but significantly better than the remaining four designs, which are in turn not fully separable from one another except for Checkerboard, which stands alone at the bottom.

Incidence-Guided Saturation Quadrants is the best or statistically tied-for-best design in every incidence configuration tested. High Incidence Focus is essentially tied with it under iid Uniform incidence (0.0800 vs. 0.0800, since an uninformative incidence surface makes “top half by incidence” close to a random half-split) but degrades sharply under Poisson incidence (MSE roughly 2.5× higher than its best-case values), because Poisson incidence is count-based and noisier at the per-cluster level, making the observed median split a less reliable proxy for the underlying risk

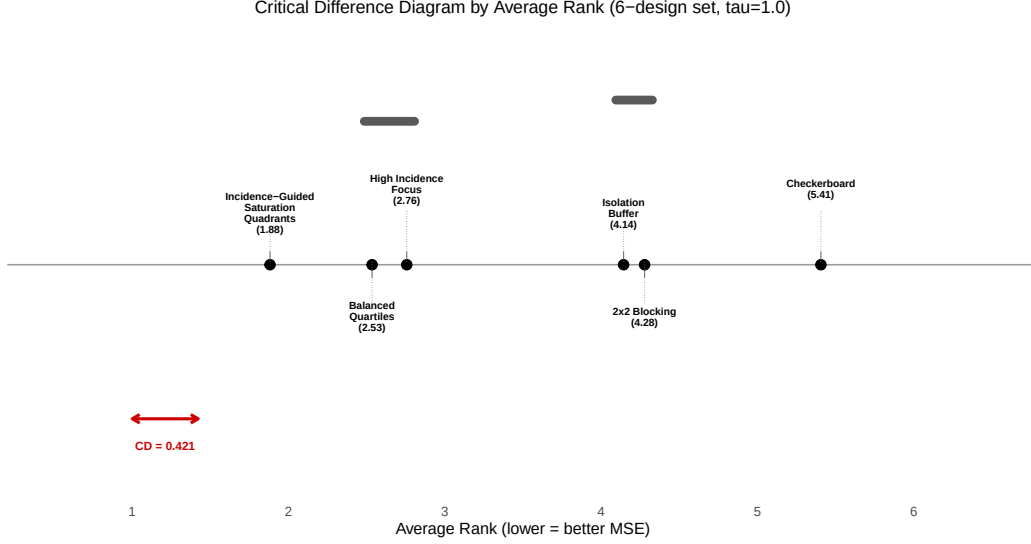


Figure S1: Critical difference diagram by average Friedman rank (six retained designs,  $\tau = 1.0$ , all incidence configurations pooled). Designs are positioned by average rank (lower = better MSE); the red arrow marks the Nemenyi critical difference ( $CD = 0.421$ ,  $\alpha = 0.05$ ). A bold bar connects designs whose average ranks do not differ significantly.

Table S5: Mean MSE by design within each incidence-generating configuration ( $\tau = 1.0$ , six retained designs). Designs are listed in the pooled best-to-worst order from Table 2 of the main text; the best design within each column is bolded.

| Design                      | iid<br>Uniform | Poisson<br>$\rho_X=0.20$ | Poisson<br>$\rho_X=0.50$ | Spatial<br>$\rho_X=0.20$ | Spatial<br>$\rho_X=0.50$ |
|-----------------------------|----------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Incidence-Guided Sat. Quad. | <b>0.0800</b>  | <b>0.0778</b>            | <b>0.0811</b>            | <b>0.0778</b>            | <b>0.0774</b>            |
| Balanced Quartiles          | 0.1000         | 0.1076                   | 0.1024                   | 0.1007                   | 0.1012                   |
| High Incidence Focus        | <b>0.0800</b>  | 0.2017                   | 0.2072                   | 0.0982                   | 0.0781                   |
| Isolation Buffer            | 0.1468         | 0.1472                   | 0.1447                   | 0.1504                   | 0.1498                   |
| 2×2 Blocking                | 0.2258         | 0.2315                   | 0.2300                   | 0.2291                   | 0.2221                   |
| Checkerboard                | 1.0586         | 0.9211                   | 0.6944                   | 0.7911                   | 0.5461                   |

surface a trial team would actually want to target. Checkerboard’s absolute MSE varies substantially by configuration (0.546 to 1.059) but remains far worse than every other design in all five configurations.

## S8 Parameter Sensitivity

Figures S3 and S4 show the same  $\rho$ -by-design pattern as a heatmap (Poisson  $\rho_X = 0.20$  representative configuration, both neighbor types), confirming that absolute MSE rises with  $\rho$  for every design but the design ranking is unchanged at every  $\rho$  level. Neither outcome spatial dependence ( $\rho$ ) nor spillover magnitude ( $\gamma$ ) meaningfully changes which design is best; both increase absolute MSE for every design (most visibly for Checkerboard, whose MSE roughly doubles from  $\gamma = 0.5$  to  $\gamma = 0.8$ : 0.710 to 0.905). The spillover exposure regime shows the largest relative effect: every design’s MSE is 2–5× higher under `control_only` spillover than under `both`, because concentrating

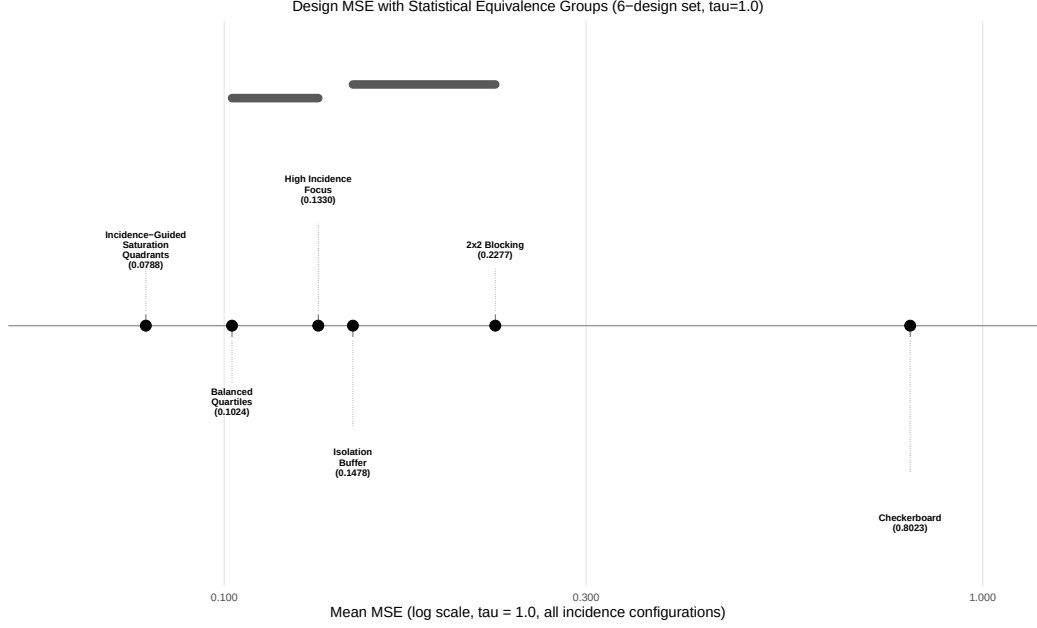


Figure S2: The same six-design comparison as Figure S1, shown by mean MSE on a log scale rather than average rank, to convey the practical magnitude of the differences alongside their statistical grouping.

Table S6: Mean MSE by design and outcome spatial dependence  $\rho$  ( $\tau = 1.0$ , six retained designs, pooled over neighbor type,  $\gamma$ , and spillover regime).

| Design                      | $\rho = 0.00$ | $\rho = 0.01$ | $\rho = 0.20$ | $\rho = 0.50$ |
|-----------------------------|---------------|---------------|---------------|---------------|
| Incidence-Guided Sat. Quad. | 0.0823        | 0.0769        | 0.0755        | 0.0807        |
| Balanced Quartiles          | 0.1003        | 0.1009        | 0.1014        | 0.1069        |
| High Incidence Focus        | 0.1097        | 0.1180        | 0.1476        | 0.1570        |
| Isolation Buffer            | 0.1480        | 0.1431        | 0.1501        | 0.1498        |
| 2×2 Blocking                | 0.2306        | 0.2219        | 0.2278        | 0.2304        |
| Checkerboard                | 0.7146        | 0.7458        | 0.9922        | 0.7566        |

spillover exposure entirely on control clusters creates a sharper asymmetry between treated and control outcomes than the direct treatment effect alone; Checkerboard is affected most severely (0.282 to 1.323, a  $4.7\times$  increase) because its rigid spatial alternation maximizes the number of control clusters adjacent to a treated cluster.

## S9 Effect-Size (Tau) Sensitivity and Rank Stability

The Friedman statistic declines monotonically as  $\tau$  increases (from 730.6 at  $\tau = 0.8$  to 557.3 at  $\tau = 3.0$ ), consistent with a compression of relative design differences at larger true effect sizes: a larger direct effect is easier for every design to estimate, narrowing the MSE gap between the best and worst designs even though the ordering itself does not change (main text Results; Figure S6). Despite this compression, every stratum remains overwhelmingly significant, and Incidence-Guided Saturation Quadrants and Balanced Quartiles remain the top two designs at every tested  $\tau$  (Fig-

Table S7: Mean MSE by design and spillover magnitude  $\gamma$  ( $\tau = 1.0$ , six retained designs, pooled over neighbor type,  $\rho$ , and spillover regime).

| Design                      | $\gamma = 0.5$ | $\gamma = 0.6$ | $\gamma = 0.7$ | $\gamma = 0.8$ |
|-----------------------------|----------------|----------------|----------------|----------------|
| Incidence-Guided Sat. Quad. | 0.0794         | 0.0792         | 0.0781         | 0.0787         |
| Balanced Quartiles          | 0.1011         | 0.1030         | 0.1049         | 0.1005         |
| High Incidence Focus        | 0.1330         | 0.1330         | 0.1331         | 0.1331         |
| Isolation Buffer            | 0.1450         | 0.1477         | 0.1493         | 0.1490         |
| 2×2 Blocking                | 0.2314         | 0.2287         | 0.2229         | 0.2278         |
| Checkerboard                | 0.7101         | 0.7648         | 0.8297         | 0.9046         |

Table S8: Mean MSE by design and spillover exposure regime ( $\tau = 1.0$ , six retained designs, pooled over neighbor type,  $\rho$ , and  $\gamma$ ).

| Design                      | Both   | Control only |
|-----------------------------|--------|--------------|
| Incidence-Guided Sat. Quad. | 0.0445 | 0.1132       |
| Balanced Quartiles          | 0.0435 | 0.1612       |
| High Incidence Focus        | 0.1043 | 0.1618       |
| Isolation Buffer            | 0.1479 | 0.1476       |
| 2×2 Blocking                | 0.0575 | 0.3979       |
| Checkerboard                | 0.2816 | 1.3230       |

ure S6), while Checkerboard’s coverage remains severely sub-nominal across the full  $\tau$  range (Figure S7), confirming that its inferential failure is a design property, not an artifact of the specific effect size used in the primary comparison.

## S10 Subgroup Robustness and Win Rate

Table order above follows median MSE (the ordering used elsewhere for the robustness profile), which differs slightly from the pooled-mean-MSE ordering used in the main text: Isolation Buffer has a higher median MSE than 2×2 Blocking but a much narrower spread (worst-case 0.206 vs. 0.821), making it the more consistent of the two even though its typical-case MSE is worse. Checkerboard’s best-case MSE (0.033, achieved only under the most favorable spatial/spillover configuration) is competitive with the top designs, but its worst-case MSE (4.81) is roughly an order of magnitude larger than any other design’s worst case, illustrating that its poor pooled average reflects catastrophic failure in a subset of configurations rather than uniformly mediocre performance.

High Incidence Focus and Incidence-Guided Saturation Quadrants together win 72.5% of all scenario blocks, consistent with Section S7’s finding that High Incidence Focus is highly competitive under favorable (low-noise/informative) incidence configurations even though its pooled-average MSE is substantially worse than Incidence-Guided Saturation Quadrants once Poisson-incidence configurations are included. Checkerboard’s 5% win rate confirms it is occasionally competitive (its best-case MSE in the table above) but is the wrong default choice for the large majority of realistic spatial and spillover conditions.

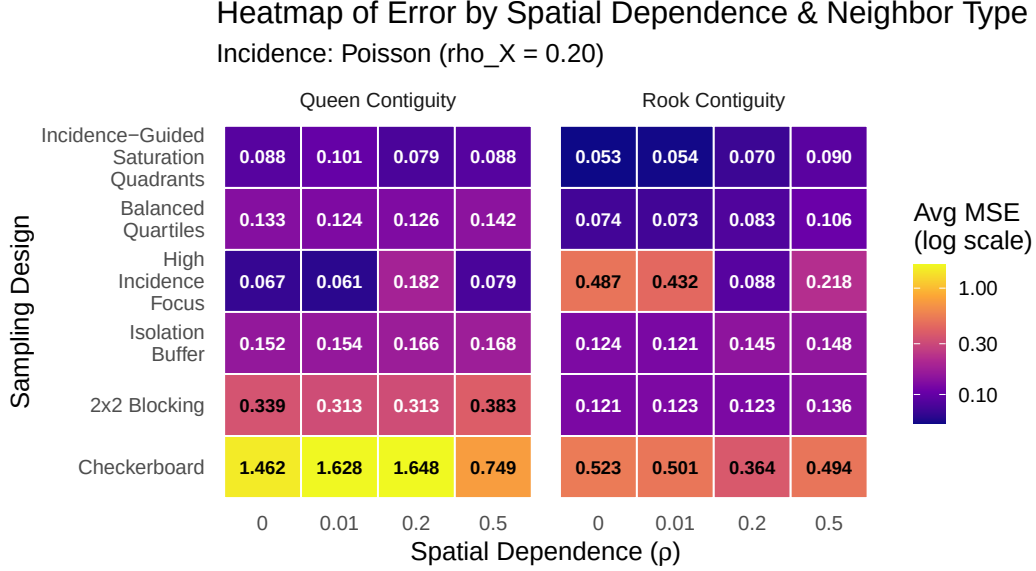


Figure S3: Mean MSE by design and outcome spatial dependence  $\rho$ , Poisson incidence ( $\rho_X = 0.20$ ,  $\tau = 1.0$ ), by neighbor type. Color scale is log-transformed to distinguish designs in the low-MSE range. Rows are ordered best (top) to worst (bottom) by pooled mean MSE.

Table S9: Friedman omnibus test for design differences, stratified by true treatment effect  $\tau$  (six retained designs, 320 blocks per stratum). All strata reject the null of equal design ranks.

| $\tau$ | $\chi^2$ (df= 5) | $p$ -value            | $N$ blocks |
|--------|------------------|-----------------------|------------|
| 0.8    | 730.60           | $< 2 \times 10^{-16}$ | 320        |
| 1.0    | 800.88           | $< 2 \times 10^{-16}$ | 320        |
| 1.5    | 705.71           | $< 2 \times 10^{-16}$ | 320        |
| 2.0    | 658.46           | $< 2 \times 10^{-16}$ | 320        |
| 3.0    | 557.29           | $< 2 \times 10^{-16}$ | 320        |

## S11 Application-Scale Illustration: Six-Design Subset

As in the main text, these numbers use a synthetic, placeholder incidence surface and must not be interpreted as a design recommendation for the real SUD application; they will be superseded once the real SUDDEN-derived North Carolina county dataset is finalized. That dataset (described in the main text’s Application section) gives county-level sudden unexpected death incidence among working-age adults (aged 18–64) for 2018–2021. It was derived from North Carolina death certificates in Habib’s unpublished master’s paper (Department of Biostatistics, University of North Carolina at Chapel Hill, 2026), and we aggregate it to the 58 service-area clusters used here.

Figures S8–S10 show the application geography itself: the 58 community-college service-area clusters used as the treatment-assignment units, the synthetic placeholder incidence surface used to inform incidence-aware designs, and the population-balanced  $k$ -means regions used by the saturation-based designs in place of the grid study’s literal  $2 \times 2$  quadrants.

*Note on design relabeling.* The application study renames two designs for the irregular, real-

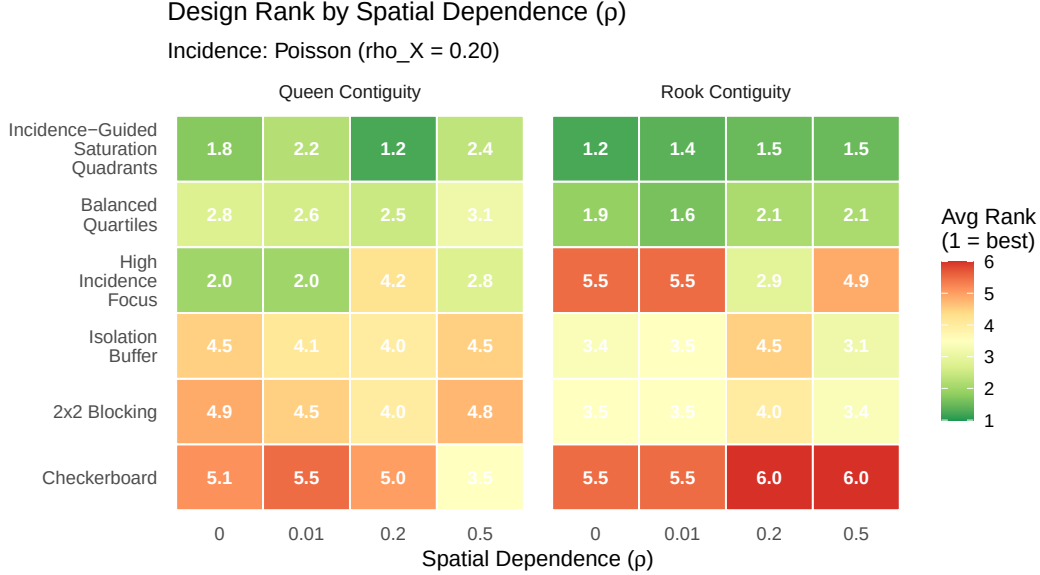


Figure S4: Average Friedman rank by design and outcome spatial dependence  $\rho$ , Poisson incidence ( $\rho_X = 0.20$ ,  $\tau = 1.0$ ), by neighbor type (rank 1 = lowest MSE = best, averaged over  $\gamma$  and spillover regime). Rows are ordered best (top) to worst (bottom).

Table S10: MSE quantiles across all 320 scenario blocks per design ( $\tau = 1.0$ , six retained designs), sorted by median MSE.

| Design                      | Best   | Q25    | Median | Q75    | Worst  |
|-----------------------------|--------|--------|--------|--------|--------|
| Incidence-Guided Sat. Quad. | 0.0344 | 0.0441 | 0.0591 | 0.1085 | 0.1894 |
| Balanced Quartiles          | 0.0332 | 0.0432 | 0.0764 | 0.1664 | 0.2743 |
| High Incidence Focus        | 0.0125 | 0.0510 | 0.0855 | 0.1658 | 0.5039 |
| 2×2 Blocking                | 0.0395 | 0.0572 | 0.1201 | 0.3111 | 0.8210 |
| Isolation Buffer            | 0.1018 | 0.1312 | 0.1464 | 0.1637 | 0.2057 |
| Checkerboard                | 0.0331 | 0.1904 | 0.4556 | 0.8167 | 4.8117 |

world community-college geography, where the grid study’s terms do not apply literally. Design ID 1 (Checkerboard on the regular  $10 \times 10$  grid) is implemented as **Block Stratified Sampling** on the irregular map — a block-stratified analog of the alternating-grid concept, since a literal checkerboard pattern has no meaning without regular grid cells. Design ID 8 (Incidence-Guided Saturation Quadrants on the grid) is implemented as **Incidence-Guided Saturation Regions**, replacing the four literal  $2 \times 2$  grid quadrants with four population-balanced  $k$ -means regions, since quadrants likewise have no meaning on an irregular map. Both are the application study’s own design names, used verbatim in the table above rather than translated back to the grid-study terminology, to avoid implying the application implementation is identical to the grid-study mechanism.

The qualitative pattern is broadly consistent with the primary grid-based results: incidence-stratified and saturation-based designs (Balanced Quartiles, Incidence-Guided Saturation Regions,  $2 \times 2$  Blocking) achieve the lowest MSE and highest power, while Block Stratified Sampling, Isolation Buffer, and High Incidence Focus are clearly worse on both dimensions. Two notable differences from the grid study:  $2 \times 2$  Blocking, which performed poorly (rank 5 of 6) on the regular grid, is

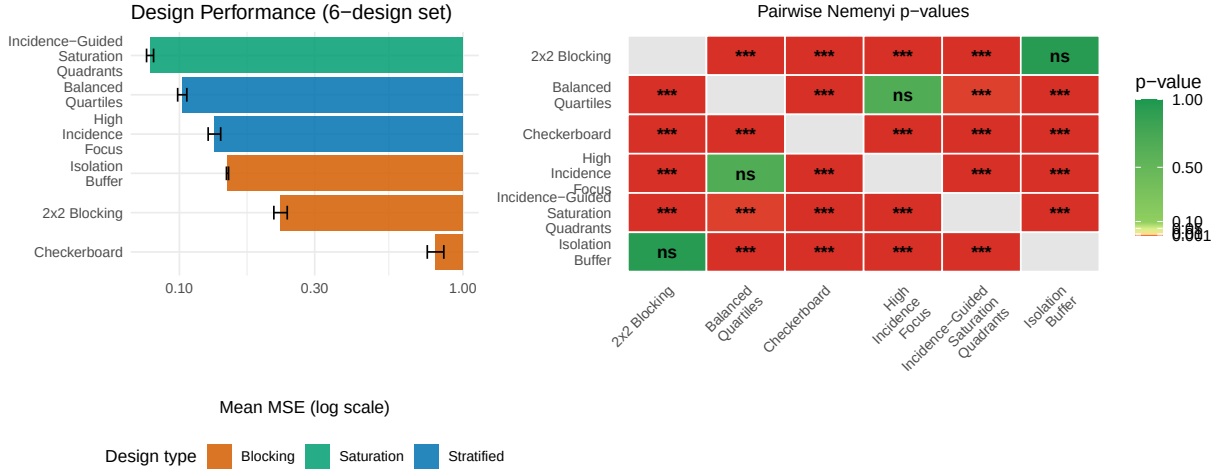


Figure S5: Left: mean MSE per design (log scale,  $\tau = 1.0$ ) with  $\pm 1$  standard error bars, colored by conceptual design group (Blocking: Checkerboard, 2 $\times$ 2 Blocking, Isolation Buffer; Stratified: High Incidence Focus, Balanced Quartiles; Saturation: Incidence-Guided Saturation Quadrants). Right: pairwise Nemenyi  $p$ -value heatmap for the six retained designs; dark red cells are statistically significant ( $p < 0.05$ ).

Table S11: Win rate: fraction of the 320 scenario blocks ( $\tau = 1.0$ ) in which each design achieves the lowest MSE among the six retained designs.

| Design                                | Win rate |
|---------------------------------------|----------|
| High Incidence Focus                  | 0.3656   |
| Incidence-Guided Saturation Quadrants | 0.3594   |
| Balanced Quartiles                    | 0.1906   |
| Checkerboard                          | 0.0500   |
| Isolation Buffer                      | 0.0250   |
| 2 $\times$ 2 Blocking                 | 0.0094   |

competitive here (rank 3), plausibly because its application-scale implementation uses local spatial blocks defined from cluster centroids, incidence, and population rather than literal  $2 \times 2$  grid cells; and High Incidence Focus, which was competitive on the grid, is one of the three worst designs here, consistent with its grid-study sensitivity to how informative the observed incidence surface is (Section S7) — the NC community-college incidence surface used for this placeholder illustration may be less informative for top-half targeting than the grid study’s iid or spatial configurations. All coverage values remain near the nominal 95% level except for Block Stratified Sampling (93.0%), which nonetheless does not show the severe under-coverage seen for Checkerboard on the regular grid (52.7%) — a difference attributable to the irregular map’s block-stratified adaptation not reproducing the same worst-case  $Z/WZ$  collinearity as the literal alternating grid pattern.

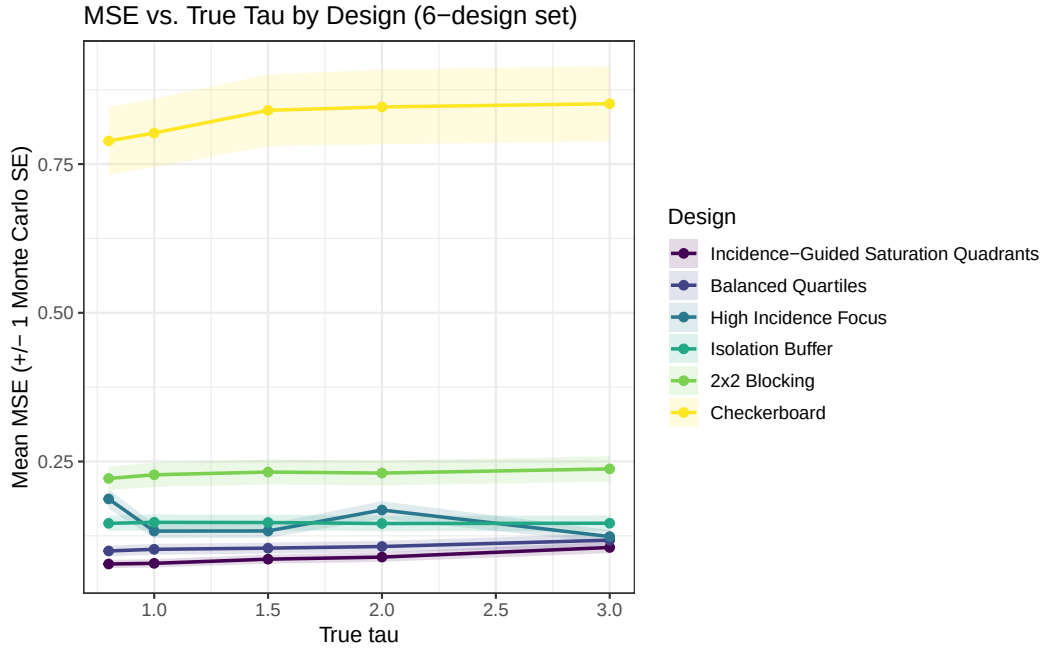


Figure S6: Mean MSE by design across the five tested treatment-effect magnitudes ( $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ ), with  $\pm 1$  Monte Carlo standard error ribbons. Legend is ordered best to worst by pooled  $\tau = 1.0$  MSE.

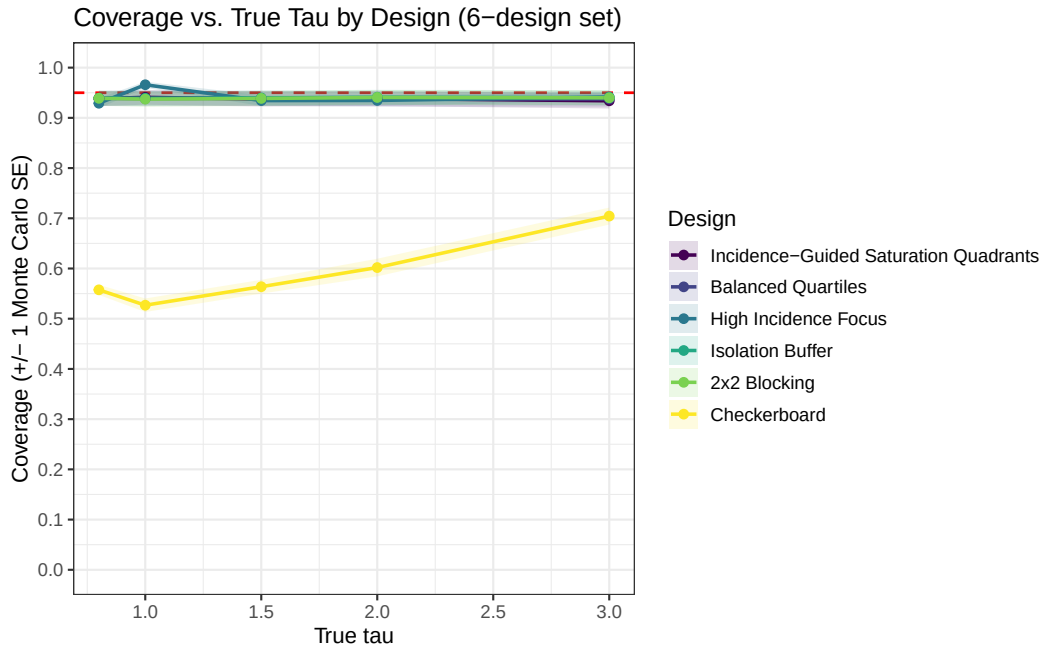


Figure S7: 95% confidence interval coverage by design across the five tested treatment-effect magnitudes, with  $\pm 1$  Monte Carlo standard error ribbons. The dashed red line marks the nominal 95% level.

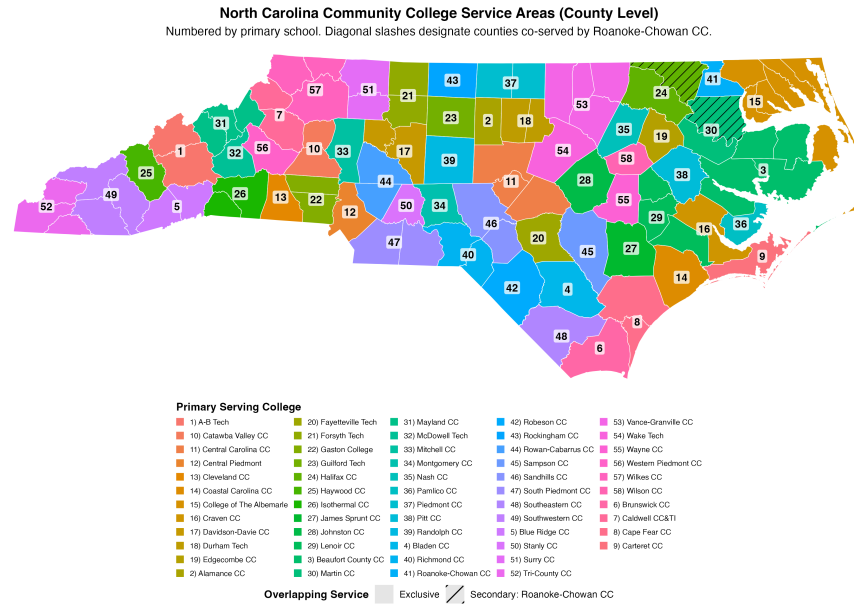


Figure S8: North Carolina Community College service-area clusters, numbered by primary serving college. This 58-cluster irregular geography replaces the grid study's regular  $10 \times 10$  lattice for the application illustration.

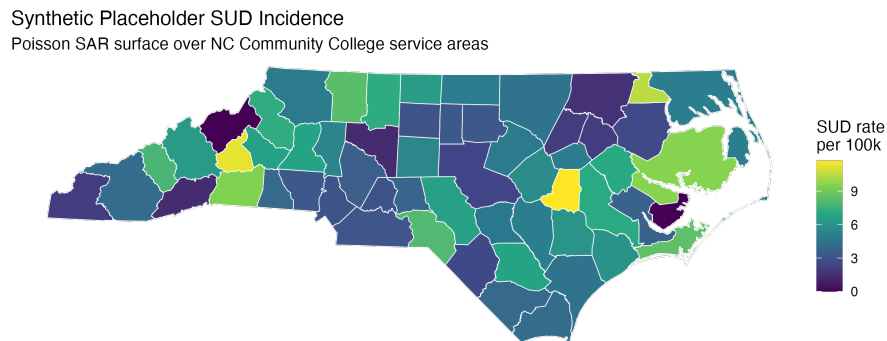


Figure S9: Synthetic placeholder SUD incidence surface (Poisson SAR process) used to inform incidence-aware designs for this illustration. This is not real surveillance data; see the placeholder caveat above.

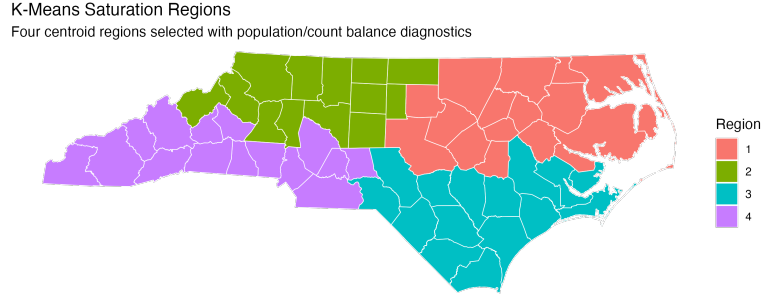


Figure S10: Population-balanced  $k$ -means regions used by Saturation Regions and Incidence-Guided Saturation Regions in place of the grid study’s literal  $2 \times 2$  quadrants (see the design-relabeling note below).

Table S12: Mean MSE, coverage, and power for the six retained designs on the 58-cluster North Carolina community-college application geography ( $\tau = 1.0$ , averaged over  $\gamma$ ,  $\rho$ , and spillover regime). Design names are the application study’s own labels (`application_designs.R`); see note below for two relabelings relative to the primary grid-based study.

| Design (application label)          | MSE    | Coverage | Power |
|-------------------------------------|--------|----------|-------|
| Balanced Quartiles                  | 0.0701 | 0.938    | 0.971 |
| Incidence-Guided Saturation Regions | 0.0778 | 0.934    | 0.966 |
| $2 \times 2$ Blocking               | 0.0782 | 0.928    | 0.961 |
| Block Stratified Sampling           | 0.2323 | 0.930    | 0.592 |
| Isolation Buffer                    | 0.2649 | 0.938    | 0.553 |
| High Incidence Focus                | 0.2686 | 0.939    | 0.535 |